Fiscal Year:	FY 2005	**Task Last Updated:**	FY 08/30/2006
PI Name:	Zeitlin, Cary J. Ph.D.		
Project Title:	Measurement of Charged Particle Interactions in Spacecraft and Planetary Habitat Shielding Materials		
Division Name:	Physical Sciences		
Program/Discipline:	RADIATION SHIELDING		
Program/Discipline--Element/Subdiscipline:	RADIATION SHIELDING--Radiation shielding		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	94720	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	98-HEDS-05
Start Date:	05/17/2000	**End Date:**	11/30/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	0		
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Grant/Contract No.:	H-31909D		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	There are two specific objectives of this project: (1) To measure nuclear fragmentation cross sections in elemental components of spacecraft and planetary habitat shielding and thus to improve the physics database that goes into the radiation transport models (2) To validate the model calculations by measurements of the charged particles produced by nuclear interactions in shielding samples This is an iterative process; at each stage, model calculations must be validated in the laboratory, and the results of the accelerator measurements fed back into model development, leading ultimately to more accurate, precise and efficient models and to shielding materials with improved radiation protection characteristics. We will complete analysis of data from a series of accelerator-based measurements in support of the development of more accurate and precise models of HZE transport in spacecraft and planetary habitat radiation shielding. NASA is committed to protecting astronauts on future exploration class missions against the deleterious effects of the radiation found in space. In particular, HZE particles in the GCR are very penetrating and potentially very damaging to cells and tissue. Due to the complex nature of the interactions of these ions, the mixture of particles in the radiation field inside a spacecraft or planetary habitat will be quite sensitive to the thickness and composition of shielding. Since it is impractical to test a large number of candidate shielding materials in flight, ground-based methods for optimizing shield performance need to be developed, and materials with desirable radiation transmission characteristics identified and evaluated. Models of radiation transport in matter can be used to rapidly evaluate many different shielding materials and configurations. However, at the present time these models are neither sufficiently accurate nor sufficiently precise, and uncertainties in the models translate into a requirement for more shielding and thus increased mission costs. Both improved source terms, which reflect the underlying physics, and validation of model, results are needed. The nuclear cross section database and charged particle transport model developed in this task is part of a NASA effort to develop new materials that will contribute to optimizing spacecraft radiation shielding properties while maintaining structural integrity and reasonable cost. This program is pertinent to any NASA endeavors involving extended-duration crewed missions, including the International Space Station and, in the longer term, possible missions to a lunar base and/or to Mars.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Data from many experiments were analyzed and the results have been made available to other NASA researchers via a Lawrence Berkeley Lab website. These results will be submitted to peer-reviewed journals for publication in the coming months. The task has been merged into a larger project concerning the physics of space radiation, which is still ongoing and consists of both a Theory Consortium and a Measurements Consortium. The work performed under this grant has become part of the work of the Measurements Consortium headed by Dr. Jack Miller of LBNL.
Bibliography Type:	Description: (Last Updated: 02/27/2004)